

Unit

1

Physics and Measurements

Teaching Periods 17

Weightage % 11



Pride of Pakistan, World's second highest peak, K-2, summit of 8611 meters.

In this unit student should be able to:

- Describe Physics.
- Describe the scope of Physics in science, technology and society.
- State SI base units, derive units and supplementary units for various measurements.
- Express derived units as products or quotients of the base units.
- State the conventions for indicating units as set in the SI units.
- Measure, using appropriate techniques, the length, mass, time temperature and electrical quantities by making use of both analogue scales and digital displays particularly, short time interval by ticker timer and by C.R.O.
- Check the homogeneity of physical equations by using dimensionality and base units.
- Derive formulae in simple cases using dimensions.
- Why all measurements contain some uncertainty.
- Distinguish between systematic errors (including zero errors) and random errors.
- Measure the diameters of few ball bearings of different sizes and estimate their volumes mention the uncertainty in each result.
- Analyze and evaluate the above experiment and suggest improvements.
- Assess the uncertainty in a derived quantity by simple addition of actual, fractional or percentage uncertainties.
- Identify dependent and independent variables.
- Draw line of best fit and error bar
- Draw extrapolation.
- Write answers with correct scientific notation, number of significant figures and units in all numerical and practical work.
- Identify that least count or resolution of a measuring instrument is the smallest increment measurable by it.
- Differentiate between precision and accuracy.
- Explain why it is important to use an instrument of smallest resolution.
- Explain the importance of increasing the number of readings in an experiment.
- Interpret the information from linear or nonlinear graphs / curves by measuring slopes and intercepts.

Measurements in physics play a crucial role in understanding and quantifying physical phenomena. They involve the determination of various properties such as length, mass, time, temperature and energy. Precise and accurate measurements provide the foundation for scientific investigations and the development of theories and models.

1.1 Scope of Physics:

What do fly birds, moving automobiles, looking blue sky and vibrating cellular phones have in common? They all follow physics or physical laws.

Birds use the difference in air pressures above and below their wings to keep them aloft. Automobiles follow the principles of mechanics and thermodynamics to transfer stored gasoline for moving tires. The sky seems blue when sunlight strikes and scatters off nitrogen and oxygen molecules in the atmosphere. Lastly, cellular phones use electronic components and the principles of electromagnetic waves to transfer energy and information from one to another phone.

1.1.1 Physics:

Physics is a branch of science that seeks to understand and describe the fundamental principles and laws governing the natural world. It encompasses the study of matter, energy, space, and time, and how they interact with each other.

The goal of physics is to develop a set of mathematical models and theories that can explain and predict the behavior of physical systems. These models are based on observations, experiments, and measurements, and are refined through a process of hypothesis testing and verification.

Physics is often divided into several sub-disciplines, each focusing on different aspects of the physical world:

Classical Mechanics: This branch deals with the motion of objects and the forces that act upon them. It includes concepts such as Newton's laws of motion, kinematics, momentum, and energy.

Electromagnetism: Electromagnetic theory explains the behavior of electric and magnetic fields, as well as their interaction with charged particles and currents. It encompasses topics like electric and magnetic forces, electromagnetic waves, and the principles underlying electricity and magnetism.

DO YOU KNOW?

Migrating birds use celestial cues and magnetic field generated by Earth's molten core to reach their destination



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Thermodynamics: Thermodynamics deals with the study of heat, energy, and their transformations. It explores concepts like temperature, entropy, energy conservation, and the behavior of gases and fluids.

Optics: Optics focuses on the properties and behavior of light. It covers topics such as reflection, refraction, diffraction, and the interaction of light with various materials and optical systems.

Quantum Mechanics: Quantum mechanics is the branch of physics that deals with the behavior of particles at the atomic and subatomic levels. It describes phenomena that classical mechanics cannot explain, such as wave-particle duality, quantum superposition, and quantum entanglement.

Relativity: Relativity theory, including both special relativity and general relativity, deals with the behavior of objects in extreme conditions, such as those involving high speed or strong gravitational field. It explores concepts like mass variation, time dilation, length contraction, and the curvature of space-time.

These are just a few examples of the many subfields within physics. Physics plays a crucial role in understanding the fundamental nature of the universe, from the microscopic world of particles to the vast scales of galaxies and the cosmos. It also underlies many technological advancements and has practical applications in various industries, including engineering, medicine, and telecommunications.

1.1.2 Scope of Physics Science, Technology and Society:

Physics plays a significant role in science, technology and society across various domains. Here are some key aspects highlighting the scope of physics in these areas:

Science:

Fundamental Laws: Physics provides fundamental laws and principles that form the basis of understanding the natural world. It explores the behavior of matter, energy, forces and their interactions, enabling scientists to develop theories and models to explain phenomena.

Advancing Knowledge: Physics drives scientific progress by pushing the boundaries of our understanding. It seeks to uncover new insights into the nature of the universe, from the microscopic realm of particles to the vast expanse of the cosmos.

Interdisciplinary Connections: Physics often intersects with other scientific disciplines, such as chemistry, biology, astronomy, and geology. It provides a framework for understanding complex systems and phenomena, facilitating interdisciplinary research and collaboration.